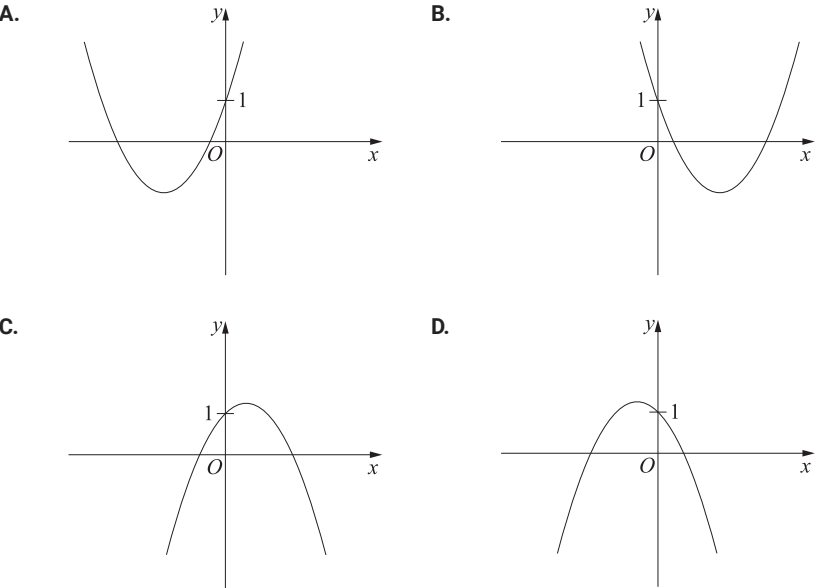


(D)

Graph (D) shows a decreasing curve in the first quadrant, starting at point (a, y) and ending at point (b, y) , where $a < b$. The curve is concave up.

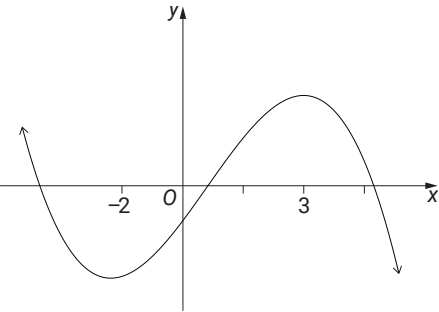
2. Calculus, 2ADV C3 2020 HSC 5 MC

Which of the following could represent the graph of $y = -x^2 + bx + 1$, where $b > 0$?



3. Calculus, 2ADV C3 2021 HSC 7 MC

The diagram shows part of $y = f(x)$ which has a local minimum at $x = -2$ and a local maximum at $x = 3$.



Which of the following shows the correct relationship between $f''(-2)$, $f(0)$ and $f'(3)$?

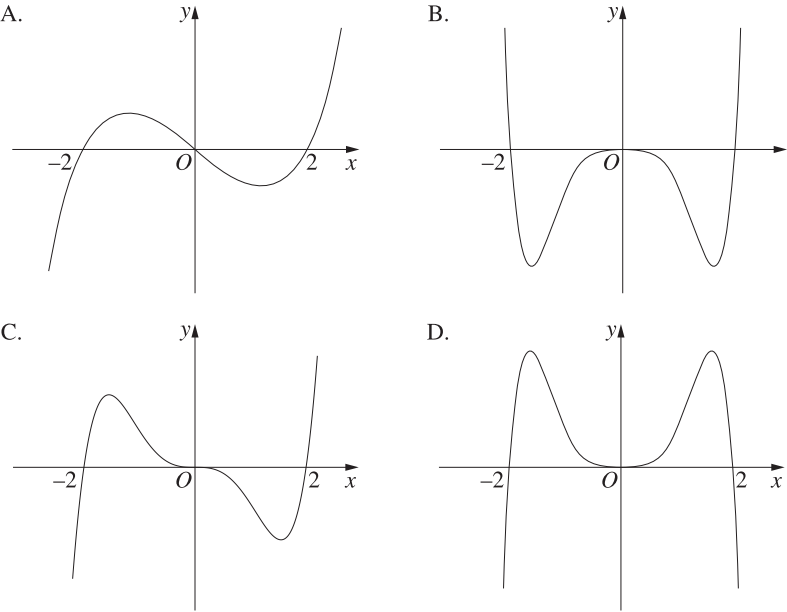
- A. $f(0) < f'(3) < f''(-2)$
- B. $f(0) < f''(-2) < f'(3)$
- C. $f''(-2) < f'(3) < f(0)$
- D. $f''(-2) < f(0) < f'(3)$

4. Calculus, 2ADV C3 2023 HSC 6 MC

The following table gives the signs of the first and second derivatives of a function $y = f(x)$ for different values of x .

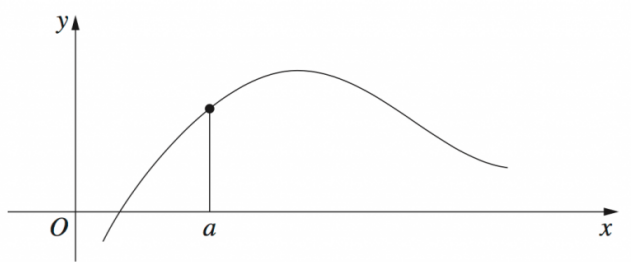
x	-2	0	2
$f'(x)$	$+$	0	$+$
$f''(x)$	$-$	0	$+$

Which of the following is a possible sketch of $y = f(x)$?



5. Calculus, 2ADV C3 2012 HSC 4 MC

The diagram shows the graph $f(x)$.

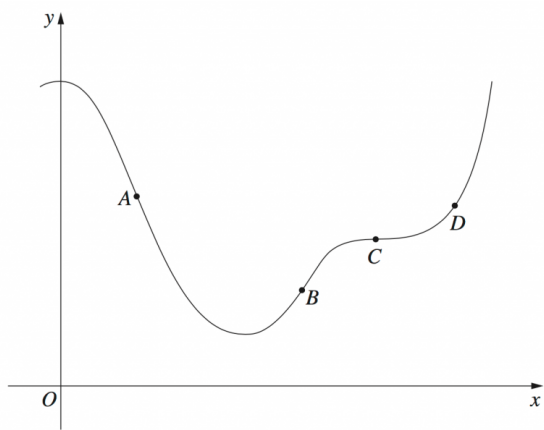


Which of the following statements is true?

- A. $f'(a) > 0$ and $f''(a) < 0$
- B. $f'(a) > 0$ and $f''(a) > 0$
- C. $f'(a) < 0$ and $f''(a) < 0$
- D. $f'(a) < 0$ and $f''(a) > 0$

6. Calculus, 2ADV C3 2013 HSC 8 MC

The diagram shows points A , B , C and D on the graph $y = f(x)$.

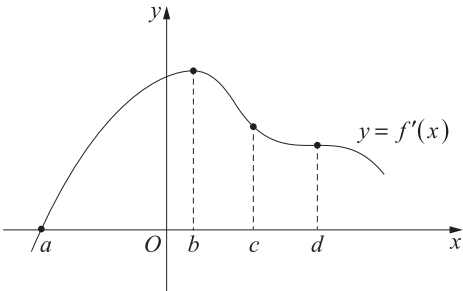


At which point is $f'(x) > 0$ and $f''(x) = 0$?

- A. A
- B. B
- C. C
- D. D

7. Calculus, 2ADV C3 2018 HSC 9 MC

The diagram shows the graph of $f'(x)$, the derivative of a function.

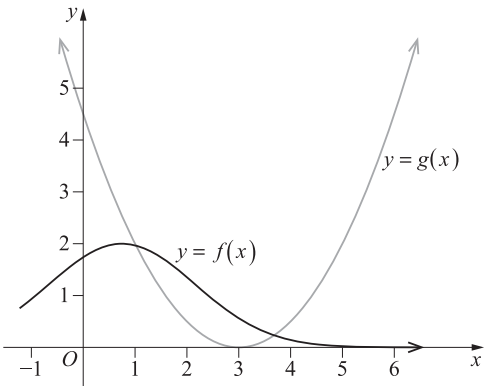


For what value of x does the graph of the function $f(x)$ have a point of inflection?

- A. $x = a$
- B. $x = b$
- C. $x = c$
- D. $x = d$

8. Calculus, 2ADV C3 2020 HSC 10 MC

The graph shows two functions $y = f(x)$ and $y = g(x)$.



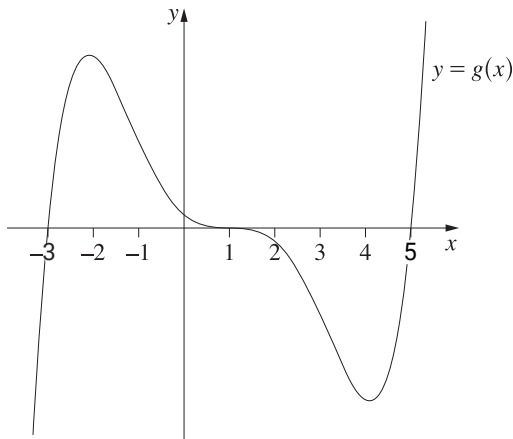
Define $h(x) = f(g(x))$.

How many stationary points does $y = h(x)$ have for $1 \leq x \leq 5$?

- A. 0
- B. 1
- C. 2
- D. 3

9. Calculus, 2ADV C3 2024 HSC 11

The graph of the function $g(x)$ is shown.

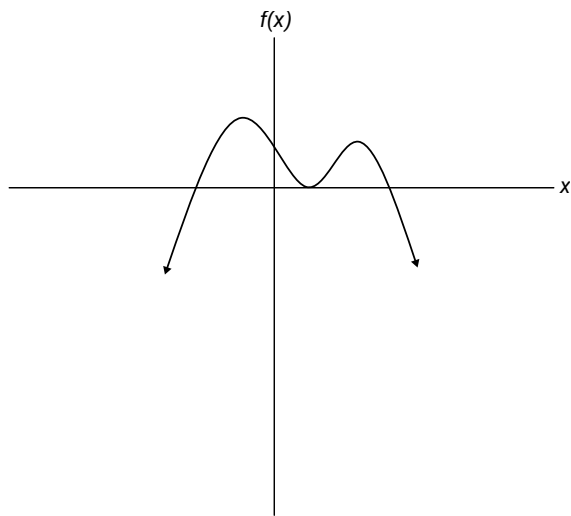


Using the graph, complete the table with the words positive, zero or negative as appropriate. (3 marks)

x -value	First derivative of $g(x)$ at x	Second derivative of $g(x)$ at x
$x = -3$		
$x = 1$		
$x = 5$		

10. Calculus, 2ADV C3 EQ-Bank 1

The graph of $f(x)$ is drawn below. Draw a sketch of $f'(x)$ on the same graph. (2 marks)



11. Calculus, 2ADV C3 2013 HSC 12a

The cubic $y = ax^3 + bx^2 + cx + d$ has a point of inflection at $x = p$.

Show that $p = -\frac{b}{3a}$. (2 marks)

12. Calculus, 2ADV C3 2021 HSC 16

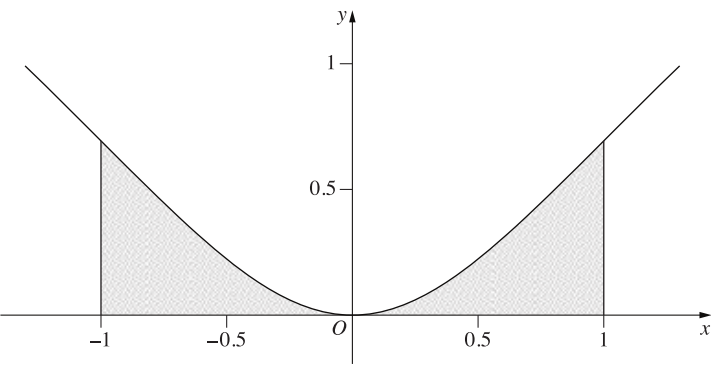
For what values of x is $f(x) = x^2 - 2x^3$ increasing? (3 marks)

13. Calculus, 2ADV C3 2014 HSC 14a

Find the coordinates of the stationary point on the graph $y = e^x - ex$, and determine its nature. (3 marks)

14. Calculus, 2ADV C4 2024 HSC 22

The graph of the function $f(x) = \ln(1 + x^2)$ is shown.



- a. Prove that $f(x)$ is concave up for $-1 < x < 1$. (3 marks)
- b. A table of function values, correct to 4 decimal places, for some x values is provided.

x	0	0.25	0.5	0.75	1
$\ln(1 + x^2)$	0	0.0606	0.2231	0.4463	0.6931

Using the function values provided and the trapezoidal rule, estimate the shaded area in the diagram. (2 marks)

- c. Is the answer to part (b) an overestimate or underestimate? Give a reason for your answer. (1 mark)

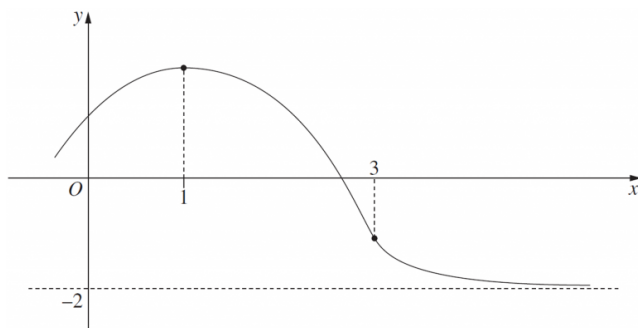
15. Calculus, 2ADV C3 2010 HSC 8d

Let $f(x) = x^3 - 3x^2 + kx + 8$, where k is a constant.

Find the values of k for which $f(x)$ is an increasing function. (2 marks)

16. Calculus, 2ADV C3 2011 HSC 9c

The graph $y = f(x)$ in the diagram has a stationary point when $x = 1$, a point of inflection when $x = 3$, and a horizontal asymptote $y = -2$.

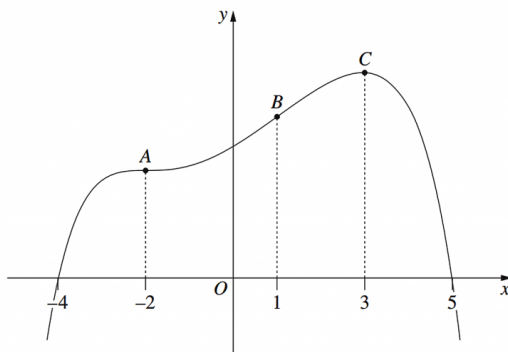


Sketch the graph $y = f'(x)$, clearly indicating its features at $x = 1$ and at $x = 3$, and the shape of the graph as $x \rightarrow \infty$. (3 marks)

17. Calculus, 2ADV C3 2014 HSC 14e

The diagram shows the graph of a function $f(x)$.

The graph has a horizontal point of inflection at A , a point of inflection at B and a maximum turning point at C .



Sketch the graph of the derivative $f'(x)$. (3 marks)

18. Calculus, 2ADV C3 2018 HSC 14c

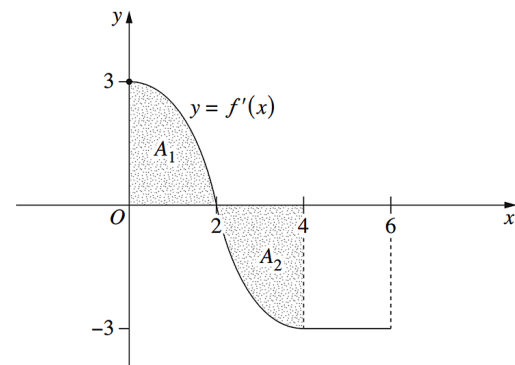
Let $f(x) = x^3 + kx^2 + 3x - 5$, where k is a constant.

Find the values of k for which $f(x)$ has NO stationary points. (3 marks)

19. Calculus, 2ADV C3 2010 HSC 9b

Let $y = f(x)$ be a function defined for $0 \leq x \leq 6$, with $f(0) = 0$.

The diagram shows the graph of the derivative of f , $y = f'(x)$.



The shaded region A_1 has area 4 square units. The shaded region A_2 has area 4 square units.

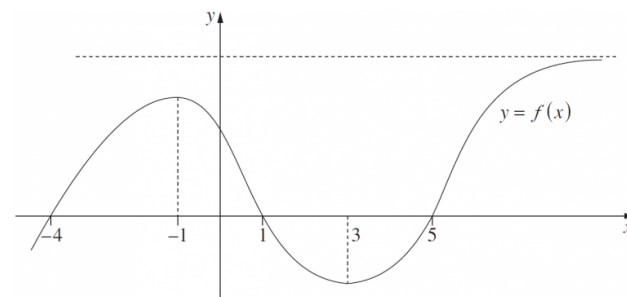
i. For which values of x is $f(x)$ increasing? (1 mark)

ii. What is the maximum value of $f(x)$? (1 mark)

iii. Find the value of $f(6)$. (1 mark)

iv. Draw a graph of $y = f(x)$ for $0 \leq x \leq 6$. (2 marks)

20. Calculus, 2ADV C3 2009 HSC 8a



The diagram shows the graph of a function $y = f(x)$.

i. For which values of x is the derivative, $f'(x)$, negative? (1 mark)

ii. What happens to $f'(x)$ for large values of x ? (1 mark)

iii. Sketch the graph $y = f'(x)$. (2 marks)

Worked Solutions

1. Calculus, 2ADV C3 2017 HSC 4 MC

Interpreting $f'(x) > 0$,
 $\Rightarrow$ gradient is always positive
Interpreting $f''(x) < 0$,
 $\Rightarrow$ Curve is concave down
 $\Rightarrow A$

2. Calculus, 2ADV C3 2020 HSC 5 MC

$y = -x^2 + bx + 1$
 $\frac{dy}{dx} = -2x + b$
S.P. occurs when $\frac{dy}{dx} = 0$:
 $-2x + b = 0$
 $x = \frac{b}{2}$
Since $b > 0$, SP occurs when $x > 0$
 $\Rightarrow C$

3. Calculus, 2ADV C3 2021 HSC 7 MC

$f''(-2) > 0$ (concave up at $x = -2$)
 $f(0) < 0$ (see graph)
 $f'(3) = 0$ (S.P.)
 $\therefore f(0) < f'(3) < f''(-2)$
 $\Rightarrow A$

4. Calculus, 2ADV C3 2023 HSC 6 MC

By elimination:
Gradient is positive at $x = -2$ and 2 (Eliminate B and D)
SP and POI at $x = 0$ (Eliminate A)
 $\Rightarrow C$

Worked Solutions

5. Calculus, 2ADV C3 2012 HSC 4 MC

At $x = a$,
 $f'(a) > 0 \therefore$ Cannot be C or D
 $f''(a) < 0$ because $f(x)$ is concave down at $x = a$
 $\therefore$ Cannot be B
 $\Rightarrow A$

6. Calculus, 2ADV C3 2013 HSC 8 MC

At A , $f'(x) < 0$
At C , $f'(x) = 0$
 $\therefore$ It cannot be A or C .
At D , $f''(x) > 0$ (concave up)
At B , $f''(x) = 0$ (concavity changes)
 $\Rightarrow B$

7. Calculus, 2ADV C3 2018 HSC 9 MC

P.I. will occur when the graph of $f'(x)$
has a turning point.
 $\therefore x = b$
 $\Rightarrow B$

8. Calculus, 2ADV C3 2020 HSC 10 MC

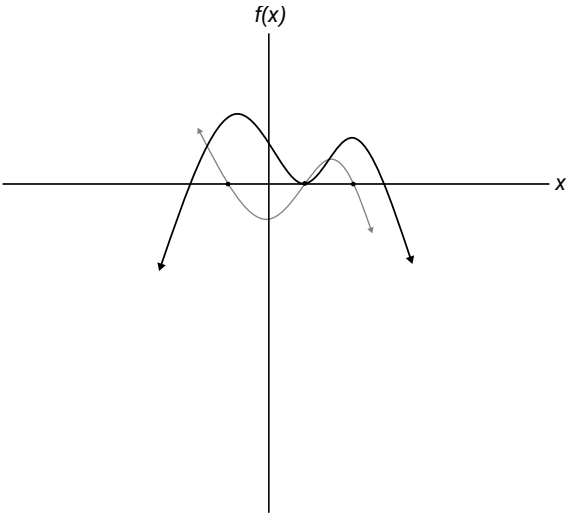
$h(x) = f(g(x))$
 $h'(x) = g'(x) \times f'(g(x))$
S.P.'s occur when $g'(x) = 0$ or $f'(g(x)) = 0$
 $g'(x) = 0$ when $x = 3$ (from graph)
 $f'(x) = 0$ when $x \approx 1$ (i.e. x -value is just under 1)
Find values of x when $g(x) \approx 1$:
By inspection, there are 2 values where
 $g(x) \approx 1, x \in [1, 5]$
 $\therefore$ There are 3 S.P.'s for $y = h(x), x \in [1, 5]$
 $\Rightarrow D$

9. Calculus, 2ADV C3 2024 HSC 11

<i>x</i> -value	First derivative of <i>g</i> (<i>x</i>) at <i>x</i>	Second derivative of <i>g</i> (<i>x</i>) at <i>x</i>
<i>x</i> = − 3	positive	negative
<i>x</i> = 1	zero	zero
<i>x</i> = 5	positive	positive

10. Calculus, 2ADV C3 EQ-Bank 1

Plot x-intercepts at the x-values of turning points.
Determine if gradient of *f*(*x*) is positive/negative to draw shape.



11. Calculus, 2ADV C3 2013 HSC 12a

Show $p = -\frac{b}{3a}$

$$y = ax^3 + bx^2 + cx + d$$
$$y' = 3ax^2 + 2bx + c$$
$$y'' = 6ax + 2b$$

Given P.I. occurs when $x = p$

$$\Rightarrow y'' = 0 \text{ when } x = p$$
$$\therefore 6ap + 2b = 0$$
$$6ap = -2b$$
$$p = -\frac{2b}{6a}$$
$$= -\frac{b}{3a} \quad \dots \text{ as required}$$

12. Calculus, 2ADV C3 2021 HSC 16

$$f(x) = x^2 - 2x^3$$
$$f'(x) = 2x - 6x^2$$

$f(x)$ is increasing when $f'(x) > 0$

$$2x - 6x^2 > 0$$
$$2x(1 - 3x) > 0$$
$$x \in \left(0, \frac{1}{3}\right)$$

13. Calculus, 2ADV C3 2014 HSC 14a

$$y = e^x - ex$$

$$\frac{dy}{dx} = e^x - e$$

$$\frac{d^2y}{dx^2} = e^x$$

S.P. when $\frac{dy}{dx} = 0$

$$e^x - e = 0$$

$$e^x = e^1$$

$$x = 1$$

At $x = 1$

$$y = e^1 - e = 0$$

$$\frac{d^2y}{dx^2} = e > 0 \Rightarrow \text{MIN}$$

$\therefore$ MINIMUM S.P. at $(1, 0)$

14. Calculus, 2ADV C4 2024 HSC 22

a. $f(x) = \ln(1 + x^2)$

$$f'(x) = \frac{2x}{1 + x^2}$$

$$f''(x) = \frac{2(1 + x^2) - 2x(2x)}{(1 + x^2)^2}$$

$$= \frac{2 + 2x^2 - 4x^2}{(1 + x^2)^2}$$

$$= \frac{2(1 - x^2)}{(1 + x^2)^2}$$

In domain $x \in (-1, 1) \Rightarrow 1 - x^2 > 0$

$(1 + x^2)^2 > 0$ for all x

$\Rightarrow f''(x) > 0$ for $x \in (-1, 1)$

$\therefore f(x)$ is concave up for $x \in (-1, 1)$

b. Total shaded area

$$\approx 2 \times \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3) + y_4]$$

$$\approx 2 \times \frac{0.25}{2} [0 + 2(0.0606 + 0.2231 + 0.4463) + 0.6931]$$

$$\approx 0.538275$$

$$\approx 0.5383 \text{ (4 d.p.)}$$

c. Trapezoidal rule assumes straight lines join points in the table.

Since the graph is concave up in the given domain, the trapezoidal rule will overestimate the area.

15. Calculus, 2ADV C3 2010 HSC 8d

$$f(x) = x^3 - 3x^2 + kx + 8$$

$$f'(x) = 3x^2 - 6x + k$$

$f(x)$ is increasing when $f'(x) > 0$

$$\Rightarrow 3x^2 - 6x + k > 0$$

$f'(x)$ is always positive

$\Rightarrow f'(x)$ is a positive definite.

i.e. when $a > 0$ and $\Delta < 0$

$$a = 3 > 0$$

$$\Delta = b^2 - 4ac$$

$$\therefore (-6)^2 - (4 \times 3 \times k) < 0$$

$$36 - 12k < 0$$

$$12k > 36$$

$$k > 3$$

$\therefore f(x)$ is increasing when $k > 3$.

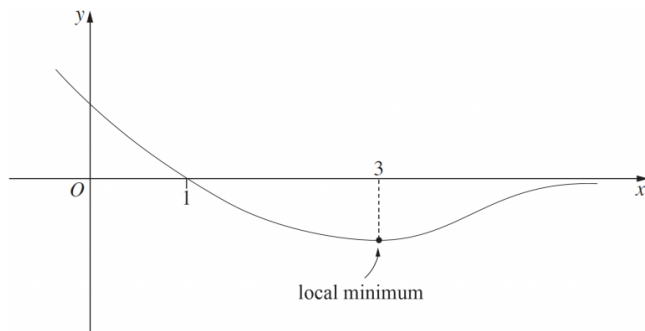
♦♦ Mean mark 28%.

MARKER'S COMMENT: The arithmetic required to solve $36 - 12k < 0$ proved the undoing of many students.

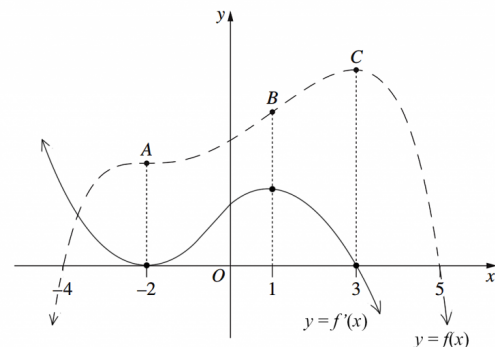
16. Calculus, 2ADV C3 2011 HSC 9c

♦ Mean mark 43%

IMPORTANT: Examiners *regularly* ask questions that require the graphing of an $f'(x)$ given the $f(x)$ graph and vice-versa. *KNOW IT!*



17. Calculus, 2ADV C3 2014 HSC 14e



18. Calculus, 2ADV C3 2018 HSC 14c

$$f(x) = x^3 + kx^2 + 3x - 5$$

$$f'(x) = 3x^2 + 2kx + 3$$

♦ Mean mark 49%.

No S.P.'s exist if $f'(x)$ has no roots,

$$\Delta < 0$$

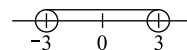
$$b^2 - 4ac < 0$$

$$(2k)^2 - 4 \times 3 \times 3 < 0$$

$$4k^2 - 36 < 0$$

$$k^2 - 9 < 0$$

$$(k-3)(k+3) < 0$$



$$\therefore -3 < k < 3$$

19. Calculus, 2ADV C3 2010 HSC 9b

- i. $f(x)$ is increasing when $f'(x) > 0$

From the graph

$f(x)$ is increasing when $0 \leq x < 2$

- ii. $f'(x) = 0$ when $x = 2$

$\therefore$ MAX at $x = 2$

$$\int_0^2 f'(x) \, dx = 4 \quad (\text{given since } A_1 = 4)$$

We also know

$$\begin{aligned} \int_0^2 f'(x) \, dx &= [f(x)]_0^2 \\ &= f(2) - f(0) \\ &= f(2) \quad (\text{since } f(0) = 0) \\ \Rightarrow f(2) &= 4 \end{aligned}$$

$\therefore$ MAX value of $f(x) = 4$

$$\begin{aligned} \text{iii. } \int_0^4 f'(x) \, dx &= A_1 - A_2 \\ &= 0 \end{aligned}$$

We also know

$$\begin{aligned} \int_0^4 f'(x) \, dx &= \int_2^4 f'(x) \, dx + \int_0^2 f'(x) \, dx \\ &= [f(x)]_2^4 + 4 \\ &= f(4) - f(2) + 4 \quad (\text{note } f(2) = 4) \\ &= f(4) \\ \Rightarrow f(4) &= 0 \end{aligned}$$

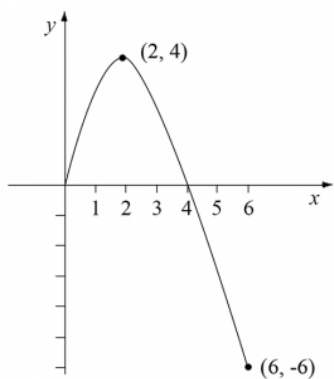
Gradient $= -3$ from $x = 4$ to $x = 6$

$$\begin{aligned} \therefore f(6) &= -3(6 - 4) \\ &= -6 \end{aligned}$$

- iv.

◆◆◆ Parts (ii) and (iii) proved particularly difficult for students with mean marks of 12% and 11% respectively.

◆◆ Mean mark 28%
EXAM TIP: Clearly identify *THE EXTREMES* when given a defined domain. In this case, the origin is obvious graphically, and the other extreme at $x = 6$, is *CLEARLY LABELLED!*

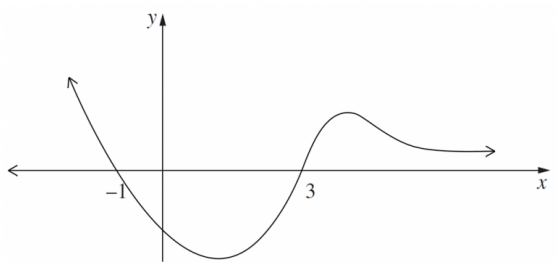


20. Calculus, 2ADV C3 2009 HSC 8a

i. $f'(x) < 0$ when
 $-1 < x < 3$

ii. As $x \rightarrow \infty$,
 $f'(x) \rightarrow 0$

iii.



♦♦ Exact data not available.

♦♦ Exact data not available.
MARKER'S COMMENT: Poorly drawn graphs with axes not labelled and inaccurate scales were common.